Exercise 3 : SQL case statement

Questions

1) Table : products

product_ID	product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

```
select product_name , price ,
  case
   when price > 1000 then 'Expensive'
   when price between 100 ~~1000~~ and 1000 then 'Mid-range'
   when price < 100 then 'Budget'
   end as '~~product~~ price_category'
  from products
```

product_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid range
Keyboard	~~45~~ 45.00	Budget
Monitor	300.00	Mid range
Mouse	25.00	Budget

2) Table : orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

```
Select customer_name , amount ,
case
when amount ≥ 1000 then 'High value'
when amount between 500 and 999.99 then 'medium'
when amount < 500 then 'low value'
end as 'order_value_category'
from orders.
```

customer_name	amount	order_value_category
Alice	150.00	low value
Bob	560.00	Midium value
Charlie	999.99	Midium value
Diana	45.50	low value
Ethan	1200.00	Hig value

3) Table : employees

emp_id	emp_name	department	salary
1	John	IT	85 000
2	Sara	HR	60 000
3	Mark	IT	75 000
4	Lucy	finance	95 000
5	Tom	HR	55 000

```
Select emp_name, department, salary,
    case
when 'IT' and salary > 80000 then 'Senior IT'
when 'HR' and salary > 55 000 then 'experienced HR'
    else staff
    end as 'position_level'
    from employees.
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60 000	experienced HR
Mark	IT	75 000	staff
Lucy	finance	95 000	staff
Tom	HR	55 000	staff

4) Table : students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

```
select student_name, score,
    case
when score ≥ 90 then 'A'
when score between 80 and 89 then 'B'
when score between 70 and 79 then 'C'
when score between 60 and 69 then 'D'
when score < 60 then 'F'
end as grade
from students
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5) Table : deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

```
Select delivery_id, delivery_time_minutes,
case
when delivery_time_minutes ≤ 30 then 'fast'
when delivery_time_minutes between 31 and 60 then
                                        'on time'
when delivery_time_minutes >60 then 'late'
End as performance
from deliveries
```

delivery_id	delivery_time_minutes	performance ~~perf~~
1	45	on time
2	80	late
3	30	fast
4	65	late
5	100	late

6) Table : tickets

ticket_id	issue_type	priority
1	login issue	1
2	server down	3
3	slow system	2
4	Email error	2
5	Password reset	1

```
Select issue_type, priority,
case
when priority = 3 then 'High'
when priority = 2 then 'Medium'
when priority = 1 then 'Low'
End as priority_label
from tickets
```

issue type	priorty	priorty_label
login issue	1	low
server down	3	High
slow system	2	Medium
Email error	2	Medium
Password reset	1	low

7) Table : ~~atted~~ attendance

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

```
Select (days_present/total_days)*100 as attendance_precentage
        student_Id,
        case
when attendance_precentage ≥ 90 then 'Excellent'
when attendance_precentage between 75 and 89 then 'Good'
when attendance_precentage < 75 then 'needs improvement'
        End as attendance_status
        from attendance
```

attendance_precentage	student_ID	attendance_status.
90%,	1	Excellent
60%	2	Good.
96%	3	Excellent
50%	4	needs improvement
100%	5	Excellent

8) Table : products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

```
Select product_id, stock_qty,
        case
when stock_qty = 0 then 'out of stock'
when stock_qty between 1 and 5 then
                        'low stock'
~~when~~ else 'In stock
        End as 'stock_status'
        from products_inventory
```

product_id	stock_qty	Stock_Status
1	5	low Stock
2	0	out of stock
3	25	In Stock
4	10	In stock
5	3	low stock

9) Table : classes

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

~~when~~ select subject, enrolled_students,
case
when enrolled_students ≥ 25 then 'large'
when enrolled_students between 10 and 25
then 'Medium'
when enrolled_students < ~~small~~ 10 then 'small'
End as class_size_category from classes

subject	enrolled_students	class_size_category
Math	30	large
English	25	large
Science	15	Medium
Art	5	small
History	20	Medium

10) Table : payments

payment_id	amount	payment_method
1	50.00	cash
2	200.00	card
3	150.00	card
4	75.00	PayPal
5	300.00	card

Select payment_id, payment_method, amount,
case
when payment_method = 'cash and
amount ≥ 200 then 'Eligible of discount'
else 'Not eligible'
End as 'discount_eligibility'
from payments

payment_id	payment_method	amount	discount_eligibility
1	card	50.00	Not eligible
2	card	200.00	Eligible of discount
3	card	150.00	Not eligible
4	PayPal	75.00	Not eligible
5	card	300.00	Eligible of discount